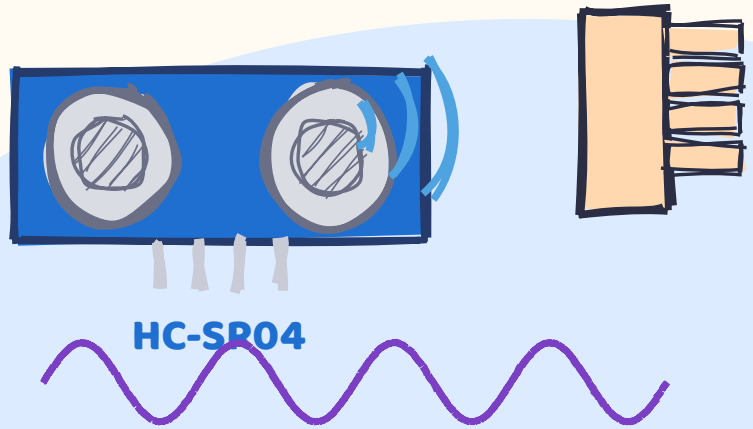
Junior Makers · Class 23

Ultrasonic Theremin
Music in the Air

Play music without touching anything

Pitch & frequency | Distance sensor | Mapping | Scales



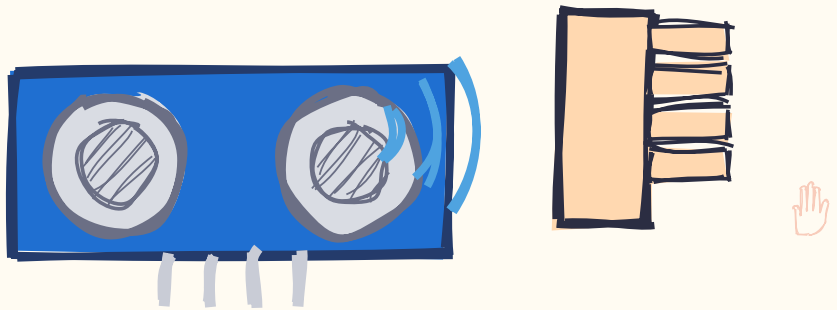
Did you know?
The theremin was invented in 1920 by Leon Theremin. It was one of the first electronic instruments ever.

Let's think...

Music without touching anything?

A magic instrument

There is a real instrument you play by waving your hands in the air. Closer = higher notes, further = lower. It sounds spooky, like a ghost singing! How could a machine 'feel' your hand without touching it?



HC-SR04

A camera?

Magic?

Echoes!

Today: you build a **theremin**: an ultrasonic sensor measures your hand and a buzzer turns distance into music.

Learning objectives

By the end of this class I can...

1	**Pitch:** explain how **frequency (Hz)** sets high and low notes	**Proof:** I can make a 262 Hz note and say what it sounds like
2	**Measure distance:** read my hand's **distance** with an ultrasonic sensor	**Proof:** my readings at 10, 20 and 30 cm
3	**Map:** **map** distance to pitch so my hand plays the buzzer	**Proof:** moving my hand closer makes the note higher
4	**Make music:** snap notes to a **musical scale** and perform a tune	**Proof:** I played the first line of a song

Objective 1 · Pitch · video

Video time!



An Untouched Instrument: The Theremin Explained

SciShow · about 4 minutes · youtu.be/KDG15-iTJLw

Watch for...

1. Who invented the theremin, and when?
2. How does the player change the **pitch**?
3. Why is it so hard to play in tune?

Scan to watch

After the video: Real theremins sense your hand with electricity. Ours will use ECHOES. Guess how!

Objective 1 · Pitch

Pitch = how fast the air shakes

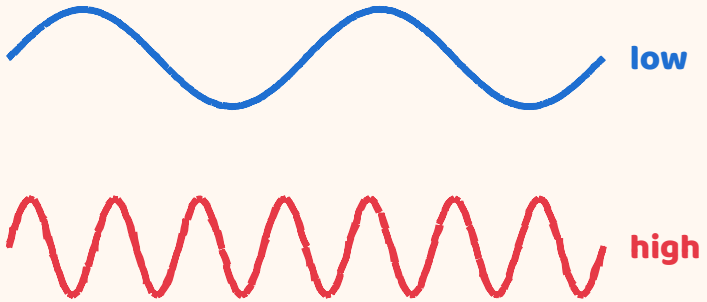
The idea

Sound is **vibration**. The number of shakes each second is the **frequency**, measured in **hertz (Hz)**.

- **Low** notes: slow shakes (a big drum: about 100 Hz)
- **High** notes: fast shakes (a whistle: about 2,000 Hz)
- Middle C on a piano: **262 Hz**

The buzzer can play any frequency we tell it.

Example: a ruler on a desk



Hold a ruler on the edge of a desk and twang it. A long overhang wobbles slowly: low sound. A short overhang wobbles fast: high sound. Same ruler, different frequency!

Objective 2 · Measure distance

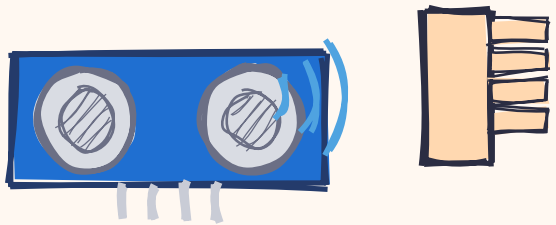
Measuring with echoes

The idea

The HC-SR04 sends out a burst of **ultrasound** (40,000 Hz: too high for us to hear). It bounces off your hand and comes back. The sensor times the echo.

Sound travels about **343 metres per second**. Short echo time = hand is close. ArduinoBlocks turns the time into **centimetres** for us.

Example: shouting in a valley



HC-SR04

Shout 'hello!' towards a mountain and you hear your echo a moment later. Further mountain = longer wait. Bats find insects in the dark with echoes, just like our sensor.

Objective 3 · Map

Map distance to pitch

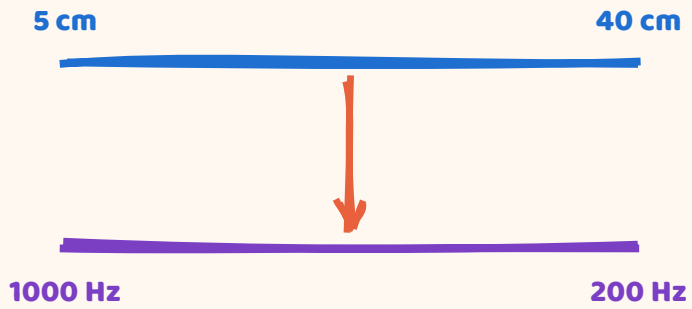
The idea

We want: hand at **5 cm** → **high note (1000 Hz)**, hand at **40 cm** → **low note (200 Hz)**. The **map** block does the maths:

map dist from 5–40 to 1000–200

Notice the second range goes **backwards** (1000 down to 200): that makes closer = higher, like a real theremin!

Example: a thermometer scale



A thermometer turns the height of a red line into a temperature. Someone decided 'this height = 0 °C, that height = 100 °C' and marked everything in between. Mapping is drawing that scale.

Objective 4 · Make music

Snap to real notes

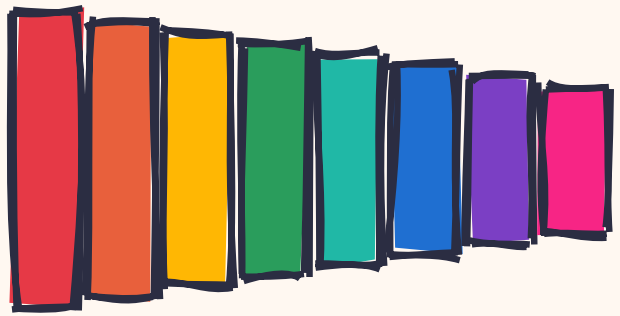
The idea

A smooth slide sounds spooky, but it's hard to play a tune. Real instruments use **notes** from a **scale**:

C 262 · D 294 · E 330 · F 349 · G 392 · A 440 · B 494 · C 523

Split the air into **8 zones**, 4 cm each. Each zone plays one note. Now it's like an invisible piano!

Example: a xylophone



A xylophone has separate bars: you can't play 'between' two notes. That's why it's easy to play a tune on it. Our zones turn the air into invisible xylophone bars.

Real world

Sensors and sound



Electronic music
Musicians invent instruments with sensors



Parking sensors
Beep faster as you get closer: distance → sound!



Medical ultrasound
Echoes make pictures of babies before birth



Accessibility
Sound-based aids help blind people sense obstacles

? A car parking sensor is basically a theremin! What does it map distance to?

Sonification
Turning data into sound helps people hear patterns, even in science data from space.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human theremin
2. Wire the theremin
3. Read your hand's distance
4. Play the air!
5. The invisible piano
6. Play a tune
7. Bug hunt

MAKE

Class game

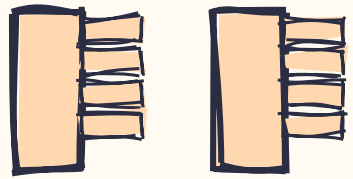
Human theremin

How to play

1. One student is the **sensor**: arm out, eyes closed.
2. Another is the **player**: they move a hand closer or further from the sensor's hand.
3. The sensor sings: **close = high 'eee'**, **far = low 'ooo'**.
4. Can the player make the sensor 'sing' a rising scale? A siren? Happy Birthday?

Then discuss

What rule did the 'sensor' follow? Could you write it as: the closer the hand, the ___ the note?



Big idea

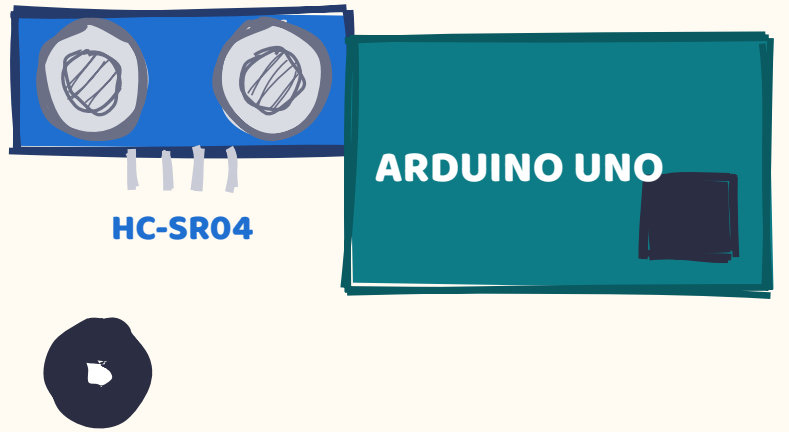
A theremin is just a rule linking distance to pitch.

MAKE

Objective 2 · Measure distance · Build step 1 of 4

Wire the theremin

Part	Arduino
HC-SR04 VCC / GND	5V / GND
HC-SR04 TRIG	pin 9
HC-SR04 ECHO	pin 10
Passive buzzer + / –	pin 8 / GND
Push button (10k to GND)	pin 2 and 5V
LED + 220 Ω	pin 12



PASSIVE buzzer!

An **active** buzzer only beeps one note. A **passive** one can play any frequency. Look for the one with a green circuit board underneath.

Objective 2 · Measure distance · Build step 2 of 4

Read your hand's distance

```
Loop
    set dist to ultrasonic distance TRIG 9 ECHO 10 (cm)
    serial print dist
    wait 100 ms
```

Hand at (ruler)	Sensor says
10 cm	
20 cm	
30 cm	
No hand	

Use a ruler

Hold a book flat in front of the sensor at exactly 10, 20 and 30 cm. Is the sensor accurate?

Strange numbers?
Very big numbers mean 'no echo'. A flat hand works better than fingers.

Objective 3 · Map · Build step 3 of 4

Play the air!

```
Loop
  set dist to ultrasonic distance (cm)
  if dist < 40 then
    set pitch to map dist from 5...40 to 1000...200
    tone pin 8 frequency pitch
    digital write pin 12 HIGH
  else
    no tone pin 8
    digital write pin 12 LOW
  wait 30 ms
```

Silence when away

The **if** stops the noise when nobody is playing. The LED shows when your hand is 'in range'.

Volume

A buzzer is LOUD in a class of 20. Agree a 'hands down' signal for quiet!

Objective 4 · Make music · Build step 4 of 4

The invisible piano

Loop

- set dist to ultrasonic distance (cm)
- set zone to round down ((dist − 5) ÷ 4)
- if zone = 0 → tone 523 (C) · if zone = 1 → tone 494 (B)
- if zone = 2 → tone 440 (A) · if zone = 3 → tone 392 (G)
- if zone = 4 → tone 349 (F) · if zone = 5 → tone 330 (E)
- if zone = 6 → tone 294 (D) · if zone = 7 → tone 262 (C)
- else → no tone

Button on pin 2: switch between SMOOTH and SCALE mode

Zone	Note
5–9 cm	C
9–13	B
13–17	A
17–21	G
21–25	F
25–29	E
29–33	D
33–37	C

Objective 4 · Make music · perform

Play a tune

Song	Notes
Twinkle Twinkle	C C G G A A G
Happy Birthday (start)	G G A G C B
Pakistan anthem (start)	teacher plays it first; find the notes by ear!
Our own tune	_ _ _ _ _ _ _

Tip

Mark the zones on paper with a ruler and lay it beside the sensor. Then you can SEE where each note is.

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Only one note or a click	active buzzer instead of passive	
Notes jump around wildly	fingers spread: use a flat hand or book	
Never silent	if dist < 40 missing	
Closer = lower	map range the wrong way round	
Distance always 0	TRIG/ECHO swapped	

Debug checklist

Print the distance and the pitch in the Serial Monitor while you play. The numbers explain the sound.

Sensors can 'see' things beside your hand. Play away from walls and tables.

Level up!

Musician challenges

Choose your challenge



Vibrato: make the note wobble up and down a little, like a singer.



Drum button: press the button for a short drum beep.



Record: store 8 notes in a list and play them back.



Duet: two theremins, two players: harmony!

Think like an engineer

Why do zones that are only 1 cm wide make the theremin almost impossible to play?

Boss level

Add a light sensor as a second 'antenna' to control volume (or speed), like a real theremin's two antennas.

Look after your ears

Sound safety

Loud = harmful

Very loud sounds for a long time can damage your hearing for ever. Keep the buzzer at arm's length, never next to someone's ear.

Headphones rule

Same for music at home: if someone next to you can hear your headphones, they're too loud.

Objective check

Can I really do it? Prove it.

01	What is frequency? Is 262 Hz higher or lower than 523 Hz?	○ ○ ○
02	Show your distance readings at 10, 20 and 30 cm.	○ ○ ○
03	Explain your map block. Why does the second range go backwards?	○ ○ ○
04	Play the first line of Twinkle Twinkle on your invisible piano.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Theremin concert: each team plays one song; the class guesses the tune.

Show what you know

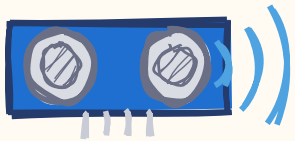
What did we learn?

- [x] Pitch is frequency: fast vibrations = high notes
- [x] Ultrasonic sensors measure distance with echoes
- [x] map turns one range of numbers into another
- [x] Snapping to a scale makes tunes easier to play
- [x] Look after your ears!

Exit ticket: your hand is at 5 cm. Will the note be high or low, and why?

At home

Twang a ruler on a table at different lengths. Which length gives the highest note?



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Next class: Processing: Interactive Drawing Machine!